

Global Warming: A Brief Overview

Global warming refers to the long-term rise in Earth's average surface temperature, driven largely by increased concentrations of greenhouse gases such as carbon dioxide and methane in the atmosphere. Since the pre-industrial era, average global temperatures have risen by roughly 1.2 degrees Celsius, with the pace of warming accelerating notably over the past four decades.

The primary driver of this warming trend is human activity, particularly the burning of fossil fuels for energy, industrial processes, and transportation. Deforestation further compounds the problem by reducing the planet's capacity to absorb carbon dioxide, while agricultural practices contribute additional methane and nitrous oxide emissions.

One of the most visible consequences of global warming is the retreat of glaciers and polar ice sheets. Satellite observations over the last few decades show significant ice mass loss in regions such as Greenland and Antarctica, contributing directly to rising sea levels that threaten coastal communities worldwide.

Extreme weather events have also become more frequent and intense as a result of warming trends. Heatwaves, prolonged droughts, and stronger tropical storms are increasingly linked by climate scientists to shifts in atmospheric and oceanic circulation patterns caused by rising global temperatures.

Efforts to address global warming center on reducing greenhouse gas emissions through renewable energy adoption, improved energy efficiency, and international agreements such as the Paris Agreement. Many scientists emphasize that limiting warming to well below 2 degrees Celsius above pre-industrial levels is essential to avoiding the most severe and irreversible impacts on ecosystems and human societies.

Note: This text is illustrative sample content generated for demonstration purposes and should not be treated as a rigorously sourced scientific reference.